

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	Total Marks:	20 marks
CO1 Statement:	Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	B. Tech IT	Date:	20/7/2026

Code of Conduct

I S.Dharmika/192421094 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S .Dharmika

Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Annexure A - QUESTIONS

1. Design a concept map that evaluates various handoff mechanisms used in cellular networks. Compare hard handoff, soft handoff, and handoff decision parameters. Assess their suitability for maintaining uninterrupted communication. Justify your evaluation using interconnected concepts.

1 HARD HANDOFF (Break-Before-Make)

PROCESS

MS disconnects from serving BS, then connects to new BS.



CHARACTERISTICS

- Only one link active at a time
- Higher chance of momentary disconnection



ADVANTAGES

- Simple to implement
- Lower complexity
- Less signaling overhead



DISADVANTAGES

- Higher call drop probability
- Interruption possible
- Poor in fast moving users



TYPICAL NETWORKS

GSM, TDMA, LTE, 5G (NSA)

WHY HANDOFF?

- User moves from one cell to another
- Signal conditions change
- To maintain call quality & uninterrupted communication



HANDOFF MECHANISMS IN CELLULAR NETWORKS

Goal: Maintain Uninterrupted Communication

2 SOFT HANDOFF (Make-Before-Break)

PROCESS

MS connects to multiple BSs simultaneously. Old link released after new link is stable.



CHARACTERISTICS

- Multiple links active
- Seamless transition
- Diversity gain



ADVANTAGES

- Higher reliability
- Better call quality
- Lower call drop probability
- Diversity gain



DISADVANTAGES

- Higher complexity
- More signaling & bandwidth required
- Costly to implement



TYPICAL NETWORKS

CDMA, WCDMA, EV-DO, 5G (SA)

SUITABLE WHEN?

Hard handoff is suitable when:

- ✓ Simplicity & low cost are priority
- ✓ Network has sufficient frequency reuse
- ✓ Interference must be minimized
- ✓ Brief interruption is acceptable
- ✓ User mobility is low to moderate

NOTE

Interference increases as reuse factor (N) decreases. Choose N wisely based on capacity & quality needs!

SIGNAL STRENGTH (RSSI/RSCP)

Higher is better

SIGNAL QUALITY (SINR/ Ec/Io)

Higher is better

MOBILE SPEED

Higher speed triggers handoff earlier

NETWORK LOAD

Prefer lightly loaded cells

BATTERY POWER

Low battery may influence decision

HYSTERESIS & TIME-TO-TRIGGER

Avoids ping-pong effect

4. EVALUATION & SUITABILITY COMPARISON

CRITERIA	HARD HANDOFF	SOFT HANDOFF	IMPACT ON UNINTERRUPTED COMMUNICATION
Call Continuity	☆☆☆☆	★★★★★	Soft handoff is far better
Call Drop Probability	High (☆☆☆☆)	Low (★★★★★)	Lower drop = better continuity
Complexity	Low (★★★★★)	High (☆☆☆☆)	Hard is simpler to implement
Bandwidth Usage	Low (★★★★★)	High (☆☆☆☆)	Hard is more bandwidth efficient
Suitability (Mobility)	Low to Moderate (☆☆☆☆)	High (★★★★★)	Soft is ideal for high mobility
Overall Suitability	☆☆☆☆	★★★★★	Soft ensures better uninterrupted communication

SUITABLE WHEN?

Soft handoff is suitable when:

- ✓ High reliability & QoS are critical
- ✓ User is in high mobility environments
- ✓ Bandwidth is available
- ✓ Interruption must be avoided
- ✓ Multiple BS coverage exists

FINAL JUDGEMENT

Soft handoff outperforms hard handoff in maintaining uninterrupted communication, especially for high mobility users and high QoS requirements. However, hard handoff remains suitable for simpler networks with limited resources.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____